

Reviewer Pack — Minimal Number of Ramanujan Matrices and Circuit Threshold F...

Entry 0c6839510976 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Number of Ramanujan Matrices and Circuit Threshold Function Inequality

Entry ID: 0c6839510976 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-06 02:06:16 UTC

1. Conjecture

Field A (mathematical branch): Ramanujan Mathematics **Field B** (complexity object): Boolean Circuit Complexity

Statement:

For any given Boolean circuit with n inputs, the minimal number of Ramanujan matrices required to represent its threshold function is upper-bounded by a constant times the logarithm of n , i.e., $\#RamanujanMatrices \leq c \log(n)$ for some constant c .

Rationale (proposer's reasoning):

The Ramanujan mathematics offers a rich algebraic structure that can potentially encode non-linear relationships within Boolean functions. The threshold function of a circuit is a crucial complexity measure related to its expressive power, and understanding the connection with Ramanujan matrices could provide insights into the hardness of computational problems.

Taxonomy category: RAMANUJANMATRICES_CIRCUITTHRESHOLD (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): db52e15f48ba068d

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For a Boolean circuit with n inputs, if the minimal number of Ramanujan matrices required to represent its threshold function is less than or equal to c times $\log(n)$ for all generated circuits and a statistically significant correlation coefficient (p -value < 0.05) between $\#RamanujanMatrices$ and $\log(n)$ is found.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.90	SAFE	SAFE
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Ramanujan matrices" AND "Boolean circuit complexity" - "threshold function inequality" AND Ramanujan's mathematics - "circuit threshold function" AND Ramanujan matrix representation

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_circuit(n):
        if n == 1:
            return ['0']
        else:
            left = generate_circuit(n // 2)
            right = generate_circuit(n - n // 2)
            return [f'({l} OR {r})' for l in left] + [f'({l} AND {r})' for l in left] + [f'NOT {l}' for l in left]

    def evaluate_circuit(circuit, input_str):
        stack = []
        for token in circuit.split():
            if token == 'OR':
                b, a = stack.pop(), stack.pop()
                stack.append('1' if a == '1' or b == '1' else '0')
            elif token == 'AND':
                b, a = stack.pop(), stack.pop()
                stack.append('1' if a == '1' and b == '1' else '0')
            elif token == 'NOT':
                a = stack.pop()
                stack.append('1' if a == '0' else '0')
            else:
                stack.append(token)
        return stack[0]

    def threshold_function(circuit, n):
        inputs = [bin(x)[2:].zfill(n) for x in range(2**n)]
```

```

        outputs = [evaluate_circuit(circuit, input_str) for input_str in inputs]
        unique_outputs = set(outputs)
        return len(unique_outputs)

def gaussian_elimination(matrix):
    m, n = len(matrix), len(matrix[0])
    rank = 0
    for i in range(n):
        pivot_row = -1
        for j in range(rank, m):
            if matrix[j][i] == '1':
                pivot_row = j
                break
        if pivot_row != -1:
            matrix[pivot_row], matrix[rank] = matrix[rank], matrix[pivot_row]
            for j in range(n):
                if i != j and matrix[rank][j] == '1':
                    for k in range(n):
                        matrix[j][k] = (int(matrix[j][k]) + int(matrix[rank][k])) % 2
            rank += 1
    return rank

def ramanujan_matrices(circuit, n):
    inputs = [bin(x)[2:].zfill(n) for x in range(2**n)]
    outputs = [evaluate_circuit(circuit, input_str) for input_str in inputs]
    matrix = [[int(outputs[j][i]) for i in range(n)] for j in range(2**n)]
    return gaussian_elimination(matrix)

n_values = [5, 10, 15, 20, 30, 40]
results = []
for n in n_values:
    circuit = generate_circuit(n)
    num_matrices = ramanujan_matrices(circuit, n)
    log_n = math.log(n)
    results.append({
        "n": n,
        "num_matrices": num_matrices,
        "log_n": log_n
    })

mean_num_matrices = sum(result["num_matrices"] for result in results) / len(results)
std_num_matrices = math.sqrt(sum((result["num_matrices"] - mean_num_matrices) ** 2 for result in results) / len(results))
correlation_coefficient = sum((result["num_matrices"] - mean_num_matrices) * (result["log_n"] - mean_log_n)) / (std_num_matrices * std_log_n)

return {
    "metric_name": "Ramanujan Matrices",
    "metric_value": mean_num_matrices,
    "instances_tested": len(results),
    "n_max": max(result["n"] for result in results),
    "conjecture_holds": correlation_coefficient > 0.95,
    "counterexample": "" if correlation_coefficient > 0.95 else f"Correlation coefficient: {correlation_coefficient}"
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 67, 71, 73, 79, 83, 89, 97]

```

```

results = []
for seed in seeds:
    result = run_trial(seed)
    print(f"TRIAL: {result}")
    results.append(result)

mean_metric_value = sum(result["metric_value"] for result in results) / len(results)
std_metric_value = math.sqrt(sum((result["metric_value"] - mean_metric_value) ** 2 for result in results) / len(results))
support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

if all(result["conjecture_holds"] for result in results):
    print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")
else:
    first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_14d85c08.py", line 105, in <module>
    result = run_trial(seed)
              ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_14d85c08.py", line 78, in run_trial
    circuit = generate_circuit(n)
              ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_14d85c08.py", line 25, in generate_circuit
    left = generate_circuit(n // 2)
           ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_14d85c08.py", line 27, in generate_circuit
    return [f'({l}) OR {r})' for l in left] + [f'({l}) AND {r})' for l in left] + [f'NOT {l})' for l in right]
           ^
NameError: name 'r' is not defined. Did you mean: 're'?

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed due to a NameError, which prevented it from producing data necessary to evaluate the conjecture. | next: Debug and correct the error in the test code to proceed with the evaluation of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	16606
2	preregistration	ollama_remote	glm4:latest	0	0	9439
3	novelty	ollama_remote	glm4:latest	0	0	8248
4	novelty	ollama_remote	glm4:latest	0	0	8934
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	20780
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9969
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13079
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15315
9	judge	ollama_remote	glm4:latest	0	0	39401

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 141771 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/0c6839510976.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/0c6839510976.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/0c6839510976.tar.gz` (if generated)